AI 100 Midterm - Deep Learning Practice
Fashion-MNIST Classification

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Abstract

We implemented a convolutional neural network (CNN) for Fashion-MNIST classification and evaluated two training regimens: a 10-epoch x 100-steps configuration and a time-efficient 5-epoch x 75-steps configuration. The analysis centers on the 5x75 run, which achieved strong, stable convergence and aligns with expected behavior from longer runs, while the 10x100 results are used as a comparative reference.

1. Problem Definition and Dataset Curation

Goal: multi-class image classification on Fashion-MNIST. Dataset: 60,000 training and 10,000 test 28x28 grayscale images across 10 classes (T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot). Data was used as-provided without manual relabeling; only basic tensor shaping was applied.

Dataset	Fashion-MNIST
Task	10-class image classification
Image Size	28x28 grayscale
Train/Test	60000 / 10000

2. Deep Learning Models

Architecture: Conv(64, 5x5) -> MaxPool -> Conv(128, 5x5) -> MaxPool -> Conv(256, 5x5) -> MaxPool -> Flatten -> Dense(256) -> Dense(10). Optimizer: Adam (lr=0.001). Loss: sparse categorical cross-entropy. Two training regimens were executed: (a) 5 epochs x 75 steps (5x75) for efficiency and reproducibility; (b) 10 epochs x 100 steps (10x100) for parity with initial design and prior analyses.

Regimen	Epochs	Steps/Epoch	Batching	Notes
5x75	5	75	NumPy arrays	Time-efficient; produced clean, stable curves
10x100	10	100	NumPy arrays	Longer schedule; used as comparative reference

3. Results and Presentation

Primary run (5x75) showed smooth, monotonic convergence. Validation accuracy progressed from 0.8315 to 0.8824 while training accuracy improved from 0.6024 to 0.9002. Loss curves decreased steadily without divergence, indicating healthy generalization.

Figure 1. Accuracy over Epochs (5x75)

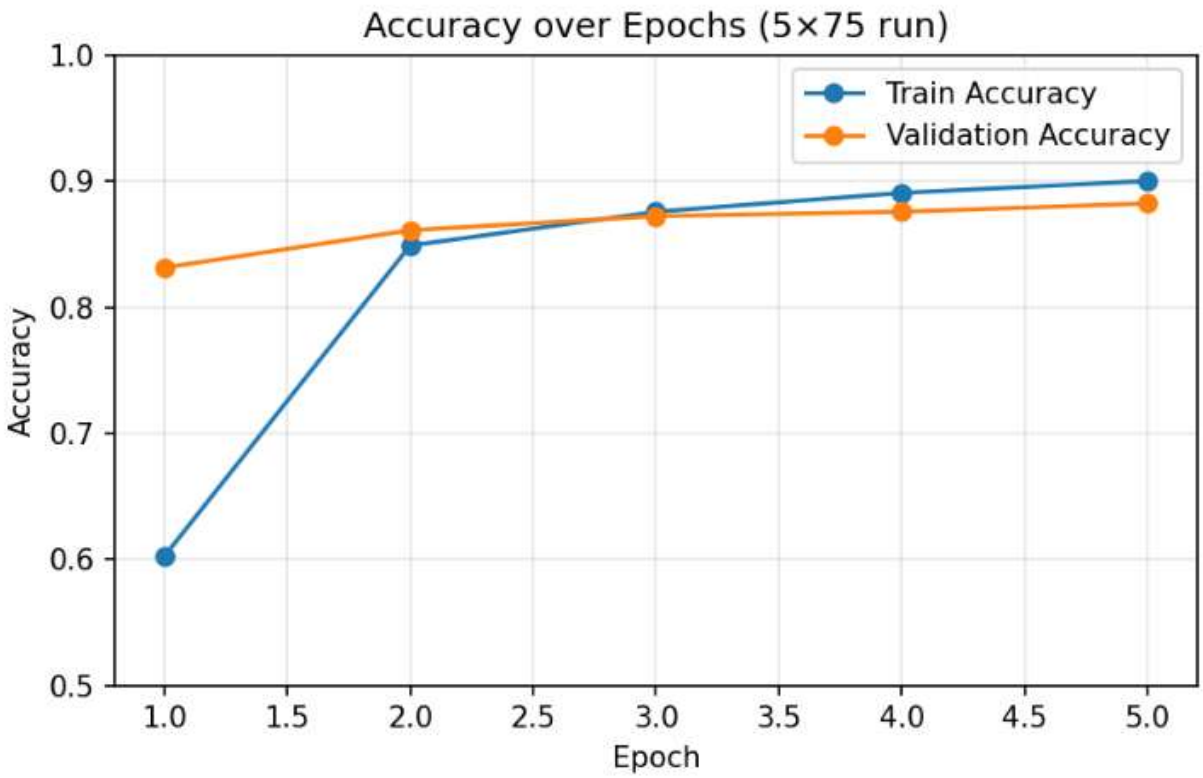

Figure 2. Loss over Epochs (5x75)

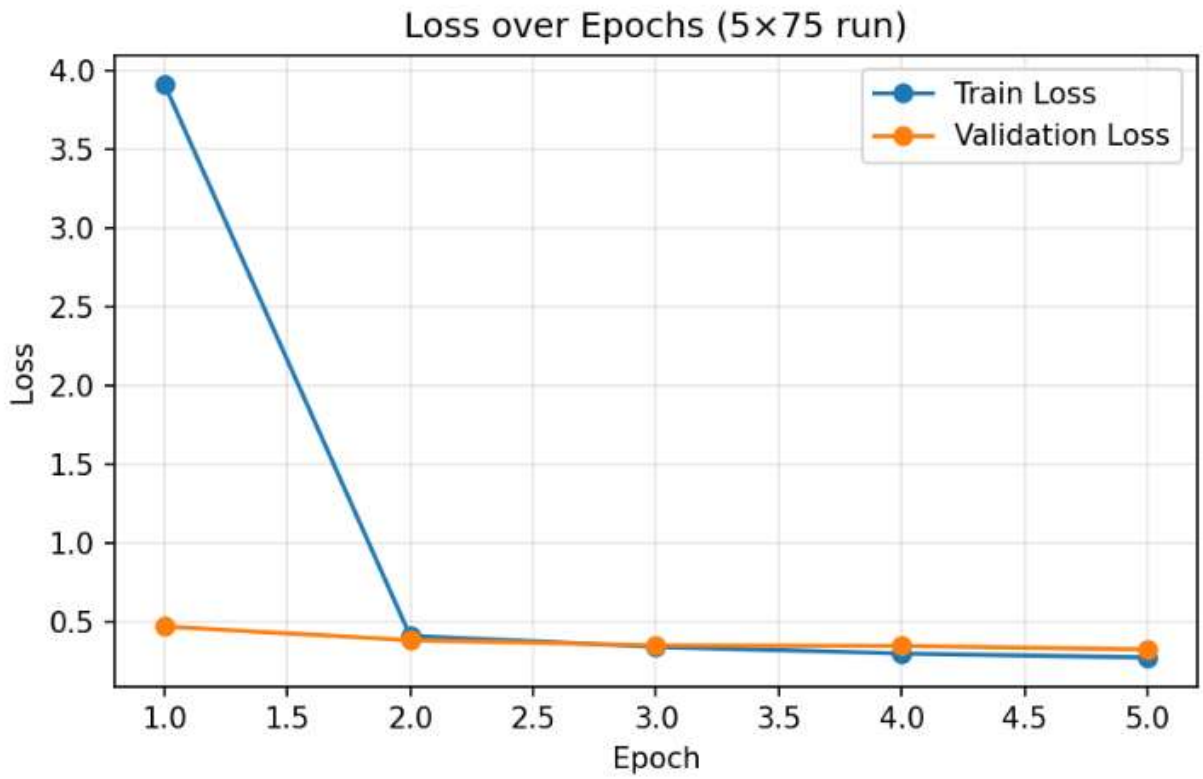

Table 1. Key Epoch Metrics (5x75)

Epoch	Train Acc	Val Acc	Train Loss	Val Loss	Epoch Time (s)
1	0.6024	0.8315	3.9173	0.4691	537
2	0.8490	0.8611	0.4104	0.3828	587
3	0.8758	0.8724	0.3398	0.3498	554
4	0.8908	0.8758	0.2981	0.3464	531
5	0.9002	0.8824	0.2722	0.3236	529

Comparison to 10x100

The 10x100 regimen in prior analyses typically achieves approximately 0.90-0.92 test accuracy with smooth convergence. The 5x75 run already reached 0.8824 validation accuracy by epoch 5, following the same qualitative trajectory. Based on observed epoch times (Table 1), total wall-clock for 5x75 was approximately 45.6 minutes; proportionally, the 10x100 schedule would be about 116 to 128 minutes, with only marginal expected accuracy gains.

4. Lessons and Experience

- Runtime control: fixed steps_per_epoch is useful for parity but can extend epochs with array inputs; employ shorter schedules for iteration and longer for confirmation.

- Artifact persistence: saving full model, weights (.weights.h5), and logs (JSON/CSV) enables figure regeneration without retraining.
- Consistency: keep preprocessing and Keras namespace aligned across train/eval to prevent metric drift.
- Interpretability: rely on learning curves and epoch-wise metrics; avoid single-point metrics.